

JAYDEN METCALFE

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EDUCATION:

Bachelor of Arts and Science, Cognitive Science

August 2020 - April 2024

McGill University, Montréal QC, Canada

- Neuroscience stream, Minor in Computer Science

Relevant Coursework: Algorithms & Data Structures, Python, Java & OOP, Probability, Software Systems, Computational Linguistics, Discrete Structures, Applied Machine Learning, Natural Language to Data Science, Linear Algebra, Deductive Logic

TECHNICAL SKILLS:

Languages: Python, TypeScript, JavaScript, Java, C, Bash (Linux), HTML, CSS, Clojure

Frameworks, Tools & Libraries: Git, React, Node.js, Express.js, Angular, PostgreSQL, Arduino, Docker, FastAPI, scikit-learn, Keras/TensorFlow, numpy, Pandas, matplotlib, REST APIs

WORK EXPERIENCE:

U16 Alpine Ski Racing Coach & Full Stack Software Developer

November 2024 - present

Whistler Mountain Ski Club, Whistler BC

- Led training sessions for 40+ athletes while coordinating with a team of 6 coaches to optimize program structure and outcomes
- Owned end-to-end development of a multi-service, full-stack training analytics web application built using React, TypeScript, Node.js, Express.js, PostgreSQL and Python
- Implemented OAuth 2.0 authentication and role-based access control for coaches, athletes, and parents
- Designed and optimized RESTful APIs and SQL queries to support filtering and comparative analytics, improving query efficiency and enabling data-driven adjustments to my team's training plan
- Containerized using Docker and Docker Compose to separate frontend, backend, and database services, improving scalability and maintainability, then deployed the application as an Amazon Web Services (AWS) EC2 instance
- Received feedback from users and implemented changes with same-day turnaround

Reinforcement Learning Researcher

August 2023 - May 2024

The Britt Lab, Montréal QC

- Automated data processing and reporting using Python and Excel, generating visualizations from thousands of data points to support research insights
- Implemented Q-learning models to simulate and analyze decision-making behaviour
- Debugged and extended Arduino code to scale for multiple experimental stages, each with variables that required fine tuning to maximize the collection of usable data points

Product Research, Design, and Development Intern

May - August 2023

Interac Corp., Toronto ON

- Prototyped an escrow-based e-transfer solution, developing RESTful API endpoints and designing a PostgreSQL database schema to enhance transaction security and reduce fraud risk, which was demoed to upper management
- Collaborated with cross-functional teams to build and update frontend features of an emulator using TypeScript and Angular within an MVC architecture, implementing UI/UX designs from Figma for client-specific product demos
- Refactored and modernized pre-existing code, improving maintainability and supporting integration with new product features

PERSONAL PROJECTS:

FIS Race Points Calculator

- Built a web application that calculates ski race scores in real time, reducing processing time from hours to seconds by integrating live timing data via API requests
- Interpreted and calculated raw data using a Python backend that spawned a Node.js subprocess, and displayed results through a dynamic front-end built with HTML, CSS and JavaScript to create a straightforward UI/UX

EXTRACURRICULARS:

McGill Varsity Alpine Ski Team

September 2020 - May 2023

- Team Captain and Varsity Council Team Representative
- Balanced a full university course load while training and competing five days a week

TEAM Mentor

January 2022 - April 2022

- Received the Tomlinson Engagement Award for Mentoring for PHYS102: Electromagnetism & Optics